

Research Note: Evidence Admission for Agent Memory

Project: Menhir / Archolith

Status: frontier research with near-term MVP implications

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Discipline transfer: forensic evidence law / admissibility doctrine

Core idea: memory systems need an admission layer, not just retrieval ranking or trust scores.

One-sentence thesis

Menhir should distinguish between evidence that merely exists, evidence that is admissible, evidence that may update a View, and evidence that may supersede current belief.

```
evidence exists
≠ evidence is admissible
≠ evidence is persuasive
≠ evidence may update current belief
≠ evidence may supersede current state
```

Current AI memory systems tend to put everything into the recall pool and rely on ranking, trust scores, or later reasoning to sort it out.

Evidence law rejects that model.

A fact-finder cannot reliably ignore unreliable evidence after exposure. Reliability must be enforced **before** the evidence reaches the decision pool.

For Menhir, this implies a new architectural layer:

```
Event
→ Foundation
→ Admission
→ Fold / Reconcile
→ View
→ Recall
```

The missing component is not another retriever.

It is a judge.

1. The Problem: Weight Cannot Fix Inadmissibility

Most AI memory systems assume bad evidence can be handled by weighting:

```
put everything in memory
→ retrieve candidates
→ rank by confidence/relevance
→ let the model decide
```

That is dangerous.

The most harmful unreliable evidence is often plausible. If it reaches the answer context, the model may use it even if it is down-ranked, contradicted, or weakly marked.

Evidence law separates two questions:

```
Is this evidence allowed to be considered?

Is this evidence persuasive?
```

Menhir should do the same.

Admission is a binary pool-membership decision.

Weight is only computed after admission.

2. Foundation-Typed Admission

Mechanism

Every memory item should carry a foundation record describing how it came to exist.

At minimum:

```
declarant
basis
personal_knowledge
asserted_at
recorded_at
```

```
source_event
process_version
```

Where `basis` is one of:

```
STATEMENT
RECORD
OBSERVATION
DERIVED
OPINION
```

The foundation is populated at intake and should not be guessed later.

Menhir rule

Perception is treated as a witness or transcriber, not as a judge.

Perception may emit:

```
STATEMENT
RECORD
OBSERVATION
```

Perception should not directly emit:

```
DERIVED total
DERIVED count
current state
conclusion
summary judgment
```

unless the output comes from an explicitly validated process.

Consequence

This reframes over-extraction.

A bad extraction is not merely “low confidence.”

It is often **inadmissible** because it has the wrong basis.

Example:

User says: "I bought an iPhone."

Bad perception emits:
PossessionCount(iPhone)=1

That is not a quoted statement. It is not a validated fold. It is a derived conclusion pretending to be a statement.

The admission gate should reject it before it enters the recallable pool.

3. Folds as Validated Expert Processes

Evidence law allows expert or technical conclusions only when the process is validated.

Menhir can adopt the same structure.

A fold is allowed to emit **DERIVED** facts because it is:

```
defined
versioned
deterministic
replayable
testable
attached to known input types
attached to known failure behavior
```

This gives every fold a "Daubert card":

```
fold_name
fold_version
input_event_types
output_view_kind
known_failure_modes
replay_hash
coverage_rule
```

Design law

```
Perception transcribes.
Folds derive.
```

```
Reconcile decides currentness.  
Recall presents admitted outputs.
```

This sharpens the existing Menhir boundary:

```
probabilistic perception  
→ deterministic event typing  
→ deterministic folds  
→ deterministic Views
```

It also explains why several LongMemEval-style failures are not perception failures.

If the answer requires a count, sum, latest state, or current value, the lawful source is a fold, not an extractor.

4. Execution-Date Supersession

Problem

Menhir has already identified a dangerous bug class:

```
arrival order accidentally becomes truth order
```

Example:

```
Event A recorded today but asserted last year  
Event B recorded yesterday but asserted last week
```

The later-recorded event should not automatically supersede the earlier-recorded one.

Evidence law distinguishes:

```
recorded_at = when the court received the document  
asserted_at = when the document was executed / the claim became true
```

A probate court does not revoke wills in filing order.

Menhir should not supersede beliefs in ingestion order.

Menhir rule

Supersession compares `asserted_at`, not `recorded_at`.

```
recorded_at controls append-only docket order
asserted_at controls temporal validity
```

If `asserted_at` is missing or vague, the event cannot supersede a dated incumbent.

It becomes contested or partially ordered.

5. Anchored Time Instead of Forced Timestamps

Humans often speak in temporal relations rather than exact dates:

```
before the migration
after the deploy
around the outage
during the flaky test incident
while we were switching ORMs
```

The wrong move is to force the LLM to invent a timestamp.

The better move is to store a temporal relation:

```
BEFORE(anchor_event)
AFTER(anchor_event)
AROUND(anchor_event)
DURING(anchor_interval)
```

This makes chronology a partial order that tightens as anchors acquire dates.

Design law

```
Vague time is not missing time.
It is constrained time.
```

This connects directly to the earlier crossdating research:

```
unknown
→ ordered
→ anchored
→ approximately dated
→ precisely dated
```

New evidence may tighten an interval.

It should not silently reverse resolved ordering without entering contested status.

6. Summary Substitution

Problem

Views and their source episodes currently risk competing in the same recall pool.

That creates double counting.

Example:

```
View:
"enrichment timeout failed 3 times"

Covered source episodes:
"timeout failed"
"timeout failed"
"timeout failed"
```

If recall sees both the View and the supporting episodes, the model may treat the summary and its sources as independent corroboration.

Evidence law solves this with summary substitution.

A summary can stand in for voluminous records, but the originals remain available for inspection.

The jury does not receive both by default.

Menhir rule

For default recall:

ACTIVE View XOR covered source episodes

Not:

ACTIVE View AND covered source episodes

Sources remain reachable through provenance drill-down.

If the View is challenged or invalidated, the substitution lifts atomically and the originals can return to the pool.

Design law

Summary with inspection right.
Not summary plus sources.

This is likely one of the cheapest near-term wins.

It directly supports the existing recall-side fix:

default recall should prefer current active Views
and suppress their covered source episodes
unless the user asks for provenance/history/debug context

7. Candidate Tier as Contested Docket

Menhir already has a CANDIDATE tier.

The evidence-law transfer changes how to think about it.

Bad design:

everything requires human approval before admission

That does not scale.

Better design:


```
rules admit ordinary evidence
candidate tier holds contested / subcritical / foundation-incomplete evidence
humans handle objections, not every item
```

So CANDIDATE is not the front door.

It is the contested-items docket.

Candidate examples

```
missing foundation
unclear asserted_at
possible derived conclusion from perception
identity unresolved
contradictory anchor relation
potential false merge
low evidence mass
```

8. Authentication and Chain of Custody

A memory system should not repeatedly mint new identities for the same object.

Evidence law assigns exhibits at intake:

```
Exhibit 14
the gun
the contract
the photograph
```

Later references cite the exhibit number.

Menhir equivalent:

```
intake-time identity resolution against registry
unresolved mentions become unidentified-pending
do not mint canonical recallable identities per episode
```

This connects to the crystallization research:

```
avoid uncontrolled nucleation
hold subcritical clusters as candidates
promote only after critical evidence mass
```

Authentication does not replace identity resolution.

It gives identity resolution a stricter intake discipline.

9. Cessation Events

Supersession and cessation are different.

A newer value supersedes an older value.

But a thing that ended did not merely receive a new value.

It ceased.

Menhir should support first-class cessation events:

```
CESSATION(subject, asserted_at, basis, source)
```

And View lifecycle states:

```
ACTIVE
SUPERSEDED
CLOSED
CANDIDATE
QUARANTINED
CROPPED
```

`CLOSED` should be distinct from `SUPERSEDED`.

Examples:

```
subscription canceled
project archived
account closed
API deprecated
```

```
server retired
person no longer in role
```

Design law

```
A thing that ceased is not merely a thing with a newer value.
```

10. “Strike That from the Record”

The most powerful question enabled by this transfer is:

```
What would we believe if this source were inadmissible?
```

Examples:

```
What would Menhir believe if this session were excluded?

What would Menhir believe if extractor version v3 were found buggy?

What would Menhir believe if this declarant/source were unreliable?

What Views change if we strike all derived facts from perception?
```

Menhir already has the expensive half:

```
replay
```

The admission layer supplies the cheap half:

```
exclude by foundation/source/process
then refold
then diff Views
```

This turns belief debugging into a deterministic query.

11. Conservation Law: The Record

The conserved quantity in this transfer is the record.

Nothing should influence belief unless it enters through the admitted record.

Menhir's version:

```
Belief is conserved from admitted evidence.
```

Folds may reorganize admitted evidence.

They may not create unsupported belief mass.

Perception may transcribe.

It may not conclude.

A recallable node whose provenance chain does not terminate in admitted raw events is a conservation violation.

Audit invariant

For any recallable View:

```
View
→ fold output
→ admitted input events
→ raw source spans / records
```

If the chain breaks, the View is inadmissible.

This turns provenance from a nice feature into an enforceable conservation law.

12. Proposed Architecture

Current architecture:

```
Event
→ Fold / Reconcile
```

```
→ View
→ Recall
```

Evidence-admission architecture:

```
Raw source
→ Event
→ Foundation record
→ Admission gate
→ Fold / Reconcile
→ View
→ Recall substitution
→ Provenance inspection
```

With phase states:

```
UNFILED
CANDIDATE
ADMITTED
ACTIVE_VIEW
SUPERSEDED
CLOSED
QUARANTINED
CROPPED
```

And core timestamps:

```
recorded_at
asserted_at
valid_interval
anchor_relation
```

13. Near-Term Implementation Targets

A. Foundation fields

Add or standardize:

```
basis
declarant
personal_knowledge
source_kind
asserted_at
recorded_at
process_version
```

B. Admission predicate

A deterministic function:

```
admit(event) -> ADMITTED | CANDIDATE | EXCLUDED
```

Initial simple rules:

```
STATEMENT with quoted source span → ADMITTED

RECORD with source/process metadata → ADMITTED

DERIVED from validated fold → ADMITTED

DERIVED from perception → CANDIDATE or EXCLUDED

OPINION from ordinary perception → EXCLUDED

missing source/provenance → CANDIDATE

missing asserted_at for supersession claim → CANDIDATE
```

C. Supersession guard

```
compare asserted_at, not recorded_at
undated cannot supersede dated
vague time stores anchor relation, not guessed timestamp
```

D. Summary substitution

For default recall:

```
if ACTIVE View covers source episodes:  
  show View  
  suppress covered source episodes  
  retain provenance drill-down
```

E. Cessation event

Add:

```
CESSATION(subject, asserted_at, foundation)
```

And lifecycle state:

```
CLOSED
```

14. Smallest Falsification Experiments

Experiment 1: Foundation Gate

Take known over-extraction examples.

Hand-classify each output:

```
quoted statement  
record  
derived conclusion  
opinion  
unsupported inference
```

Apply admission rules.

Measure whether bad emissions are disproportionately rejected compared to good emissions.

Falsifier

If good and bad outputs have the same foundation distribution, foundation gating does not cut the problem.

Experiment 2: Execution-Date Supersession

Use a known state-change sequence:

```
25 → 20 → 25
```

Replay all arrival permutations with `asserted_at` attached.

Current View must equal the temporally latest asserted state in every permutation.

Falsifier

If the guard fails under permutation, the ordering model is insufficient.

Also measure how often real state-change events contain recoverable asserted time.

If too few do, anchor relations become necessary.

Experiment 3: Summary Substitution

On queries where an ACTIVE View ranks first:

```
filter covered source episodes from candidate pool
```

Measure:

```
answer correctness  
rank contamination  
token footprint  
provenance availability
```

Falsifier

If correctness drops, Views are under-covering their source material and substitution would hide useful truth.

15. Risks

Over-exclusion

A strict admission gate may hide useful evidence.

Mitigation:

```
CANDIDATE docket  
residual exception path  
provenance inspection mode  
debug/historical lens
```

Foundation fabrication

Perception may mislabel derived conclusions as statements.

Mitigation:

```
span verifier  
spot audits  
basis-shape checks  
extractor-version reliability views
```

Summary under-coverage

A View may not actually summarize all relevant source details.

Mitigation:

```
coverage metadata  
inspection right  
atomic unsubstitution on challenge  
D0 / answer-cost regression checks
```

Too much legal complexity

Evidence law is huge.

Menhir should not import the whole system.

Only keep three mechanisms:

```
foundation-typed admission
execution-date supersession
summary substitution
```

16. What This Removes

If successful, the admission layer could simplify or replace:

```
trust-score-first filtering
human approval as default gate
ad hoc over-extraction cleanup
arrival-order supersession
Views competing with their own sources
derived facts emitted directly from perception
```

It does not replace:

```
events
provenance
folds
Views
replay
retrieval
```

It makes them stricter.

17. Recommendation

This transfer is unusually practical.

Unlike some frontier crosses, it maps directly onto existing Menhir problems:

```
over-extraction
arrival-order supersession
candidate tier semantics
View/source double-counting
```

belief-gate activation
derived fact authority

The strongest MVP candidate is **summary substitution**, because it is small, measurable, and directly addresses View/source competition.

The strongest architectural candidate is **foundation-typed admission**, because it clarifies the boundary between perception and lawful derivation.

The most urgent correctness fix is **execution-date supersession**, because arrival order should never determine current truth.

18. Final Principle

Menhir already has the courthouse:

append-only docket
transcript
evidence exhibits
provenance
replay
derived summaries

What it lacks is the judge.

The judge is not an LLM.

The judge is a deterministic admission layer that decides what evidence is allowed to enter the decision pool before recall ever sees it.